VP11 — Properties, traceability and model gaps

Revision: `b3011e73ee1eece4b0df6861ea14435f312f5e54fcccec544f2b0ceebfe7012e`.

Declarations and all facts with source positions; support boundaries are retained.

Snapshot coverage

Counts describe this checked source snapshot, not implementation coverage or verification evidence. This inspection index contains no combined architecture diagram.

Inventory	Count
Declarations	47
Facts	87
Explicit properties	29
Reported model gaps	7

Declaration kinds

Kind / verb	Count
activity	4
actor	2
capability	2
channel	2
container	2
contract	5

database	2
datagram	1
dataset	2
encoding	2
feature	1
field	6
functionality	1
interface	2
message	2
mode	3
scenario	1
unit	2
usecase	3
variant	2

Fact verbs

Kind / verb	Count
addresses	1
allocated-to	2
consumes	2
contains	1
contributes-to	1
defines	2
delivers	1

depends-on	1
encodes	2
from	1
has	29
has-field	6
holds	2
owns	3
permits	2
places	6
projects	1
provides	1
pursues	2
realizes	1
refines	2
requires	2
runs-in	1
step	1
supports	1
upholds	7
uses	6

Explicit properties

Property values, completeness and implementation-status are source assertions, not proof of implemented behavior or owner acceptance. Missing properties are unspecified, with no inferred defaults.

- ABuild: implementation-status = implemented. Source fact: f0002.
- APlan: implementation-status = planned. Source fact: f0004.
- AVerify: implementation-status = verified. Source fact: f0006.
- ChannelBody: completeness = closed. Source fact: f0009.
- ChannelEvents: completeness = closed. Source fact: f0011.
- ChannelNotice: message-kind = event. Source fact: f0014.
- ChannelUpdate: message-kind = event. Source fact: f0016.
- ChannelValue: presence = required. Source fact: f0018.
- ChannelValue: value-type = text. Source fact: f0019.
- DataRecord: completeness = closed. Source fact: f0020.
- DataValue: presence = required. Source fact: f0022.
- DataValue: value-type = text. Source fact: f0023.
- Display: repeatability = deterministic. Source fact: f0038.
- Display: state-retention = stateful. Source fact: f0039.
- Example: completeness = open. Source fact: f0042.
- PacketBig: bit-order = most-significant-first. Source fact: f0050.
- PacketBig: byte-order = big-endian. Source fact: f0051.
- PacketFlag: presence = required. Source fact: f0057.
- PacketFlag: value-type = boolean. Source fact: f0058.
- PacketLittle: bit-order = least-significant-first. Source fact: f0060.
- PacketLittle: byte-order = little-endian. Source fact: f0061.
- PacketNumber: presence = required. Source fact: f0065.
- PacketNumber: value-type = signed. Source fact: f0066.
- PacketPayload: presence = required. Source fact: f0067.
- PacketPayload: value-type = bytes. Source fact: f0068.
- UnusedShape: completeness = open. Source fact: f0077.
- WireShape: completeness = closed. Source fact: f0082.
- WireValue: presence = optional. Source fact: f0084.

- WireValue: value-type = bytes. Source fact: f0085.

Reported model gaps

These are observations from the producer's current gap rules across viewpoints, not an exhaustive requirements audit. A missing modeled relation need not be a missing implementation or a required feature. An absence has no source statement; links identify declared context only. No reported gaps does not establish model completeness.

VP01 — Use cases and traceability

- NO_CONTRIBUTION — Export: No direct or Feature-mediated Functionality contributions.
- NO_ACTOR — Isolated: No Actor pursues this use case.
- NO_CONTRIBUTION — Isolated: No direct or Feature-mediated Functionality contributions.
- NO_GOAL — Guest: Actor has no pursues relation.

VP08 — Channel contracts and sequences

- INCOMPLETE_PARTICIPATION — Bus; mode Offline: Incomplete participation for ChannelUpdate.
- NO_PARTICIPANTS — IdleBus: No explicit participation.

VP10 — Datagram encoding and packets

- NO_ENCODING — WireVariant: No fixed wire layout; a logical variant does not define a packet.

Declaration register

Every declaration is retained, including isolated definitions. Source links identify exact spans in the original checked files.

- ABuild (activity) — Overview.design, line 2.

- AIsolated (activity) — Overview.design, line 3.
- APlan (activity) — Overview.design, line 4.
- AVerify (activity) — Overview.design, line 5.
- Api (interface) — Overview.design, line 6.
- Browse (usecase) — Overview.design, line 7.
- Bus (channel) — Overview.design, line 8.
- ChannelBody (contract) — Overview.design, line 9.
- ChannelEvents (contract) — Overview.design, line 10.
- ChannelNotice (message) — Overview.design, line 11.
- ChannelUpdate (message) — Overview.design, line 12.
- ChannelValue (field) — Overview.design, line 13.
- DataRecord (contract) — Overview.design, line 14.
- DataValue (field) — Overview.design, line 15.
- Delivery (capability) — Overview.design, line 16.
- Desktop (unit) — Overview.design, line 17.
- Display (functionality) — Overview.design, line 18.
- Example (scenario) — Overview.design, line 19.
- Export (usecase) — Overview.design, line 20.
- Guest (actor) — Overview.design, line 21.
- IdleBus (channel) — Overview.design, line 22.
- Isolated (usecase) — Overview.design, line 23.
- Local (container) — Overview.design, line 24.
- MemoryData (dataset) — Overview.design, line 25.
- Navigation (feature) — Overview.design, line 26.
- Offline (mode) — Overview.design, line 27.
- Online (mode) — Overview.design, line 28.
- PacketBig (encoding) — Overview.design, line 29.
- PacketBody (variant) — Overview.design, line 30.

- PacketFlag (field) — Overview.design, line 31.
- PacketLittle (encoding) — Overview.design, line 32.
- PacketNumber (field) — Overview.design, line 33.
- PacketPayload (field) — Overview.design, line 34.
- PersistentStore (database) — Overview.design, line 35.
- Quiet (mode) — Overview.design, line 36.
- Reader (actor) — Overview.design, line 37.
- Remote (container) — Overview.design, line 38.
- SparePort (interface) — Overview.design, line 39.
- Standalone (unit) — Overview.design, line 40.
- StoredData (dataset) — Overview.design, line 41.
- Unconnected (capability) — Overview.design, line 42.
- UnusedShape (contract) — Overview.design, line 43.
- UnusedStore (database) — Overview.design, line 44.
- WireFamily (datagram) — Overview.design, line 45.
- WireShape (contract) — Overview.design, line 46.
- WireValue (field) — Overview.design, line 47.
- WireVariant (variant) — Overview.design, line 48.

Complete fact registry

All statements are preserved, including mode, ternary relation, sequence and placement arguments. Source traceability does not prove native code conformance.

- f0000 — ABuild addresses Display. (Overview.design, line 49).
- f0001 — ABuild delivers Navigation. (Overview.design, line 50).
- f0002 — ABuild has implementation-status = implemented. (Overview.design, line 51).
- f0003 — ABuild refines APlan. (Overview.design, line 52).
- f0004 — APlan has implementation-status = planned. (Overview.design, line 53).
- f0005 — AVerify depends-on ABuild. (Overview.design, line 54).

- f0006 — AVerify has implementation-status = verified. (Overview.design, line 55).
- f0007 — AVerify refines APlan. (Overview.design, line 56).
- f0008 — Bus upholds ChannelEvents. (Overview.design, line 57).
- f0009 — ChannelBody has completeness = closed. (Overview.design, line 58).
- f0010 — ChannelBody has-field ChannelValue. (Overview.design, line 59).
- f0011 — ChannelEvents has completeness = closed. (Overview.design, line 60).
- f0012 — ChannelEvents permits ChannelNotice. (Overview.design, line 61).
- f0013 — ChannelEvents permits ChannelUpdate. (Overview.design, line 62).
- f0014 — ChannelNotice has message-kind = event. (Overview.design, line 63).
- f0015 — ChannelNotice upholds ChannelBody. (Overview.design, line 64).
- f0016 — ChannelUpdate has message-kind = event. (Overview.design, line 65).
- f0017 — ChannelUpdate upholds ChannelBody. (Overview.design, line 66).
- f0018 — ChannelValue has presence = required. (Overview.design, line 67).
- f0019 — ChannelValue has value-type = text. (Overview.design, line 68).
- f0020 — DataRecord has completeness = closed. (Overview.design, line 69).
- f0021 — DataRecord has-field DataValue. (Overview.design, line 70).
- f0022 — DataValue has presence = required. (Overview.design, line 71).
- f0023 — DataValue has value-type = text. (Overview.design, line 72).
- f0024 — Delivery requires Api in mode Offline. (Overview.design, line 73).
- f0025 — Delivery requires SparePort in mode Online. (Overview.design, line 74).
- f0026 — Desktop consumes Api. (Overview.design, line 75).
- f0027 — Desktop holds MemoryData. (Overview.design, line 76).
- f0028 — Desktop owns Display. (Overview.design, line 77).
- f0029 — Desktop owns PersistentStore. (Overview.design, line 78).
- f0030 — Desktop owns UnusedStore. (Overview.design, line 79).
- f0031 — Desktop provides Delivery. (Overview.design, line 80).
- f0032 — Desktop uses Bus as receiver of ChannelNotice in mode Offline. (Overview.design, line 81).

- f0033 — Desktop uses Bus as sender of ChannelNotice in mode Online. (Overview.design, line 82).
- f0034 — Desktop uses Bus as sender of ChannelUpdate in mode Online. (Overview.design, line 83).
- f0035 — Display allocated-to Local in mode Offline. (Overview.design, line 84).
- f0036 — Display allocated-to Remote in mode Online. (Overview.design, line 85).
- f0037 — Display contributes-to Navigation. (Overview.design, line 86).
- f0038 — Display has repeatability = deterministic. (Overview.design, line 87).
- f0039 — Display has state-retention = stateful. (Overview.design, line 88).
- f0040 — Display projects StoredData into WireFamily. (Overview.design, line 89).
- f0041 — Display realizes Delivery. (Overview.design, line 90).
- f0042 — Example has completeness = open. (Overview.design, line 91).
- f0043 — Example runs-in Online. (Overview.design, line 92).
- f0044 — Example step 1 sends ChannelNotice from Desktop to Remote via Bus. (Overview.design, line 93).
- f0045 — IdleBus upholds ChannelEvents. (Overview.design, line 94).
- f0046 — Local contains Desktop. (Overview.design, line 95).
- f0047 — MemoryData upholds DataRecord. (Overview.design, line 96).
- f0048 — Navigation supports Browse. (Overview.design, line 97).
- f0049 — PacketBig encodes PacketBody. (Overview.design, line 98).
- f0050 — PacketBig has bit-order = most-significant-first. (Overview.design, line 99).
- f0051 — PacketBig has byte-order = big-endian. (Overview.design, line 100).
- f0052 — PacketBig places PacketFlag at 0 bits 1. (Overview.design, line 101).
- f0053 — PacketBig places PacketNumber at 1 bits 12. (Overview.design, line 102).
- f0054 — PacketBig places PacketPayload at 13 bits 16. (Overview.design, line 103).
- f0055 — PacketBody has-field PacketNumber. (Overview.design, line 104).
- f0056 — PacketBody has-field PacketPayload. (Overview.design, line 105).

- f0057 — PacketFlag has presence = required. (Overview.design, line 106).
- f0058 — PacketFlag has value-type = boolean. (Overview.design, line 107).
- f0059 — PacketLittle encodes PacketBody. (Overview.design, line 108).
- f0060 — PacketLittle has bit-order = least-significant-first. (Overview.design, line 109).
- f0061 — PacketLittle has byte-order = little-endian. (Overview.design, line 110).
- f0062 — PacketLittle places PacketFlag at 0 bits 1. (Overview.design, line 111).
- f0063 — PacketLittle places PacketNumber at 1 bits 12. (Overview.design, line 112).
- f0064 — PacketLittle places PacketPayload at 13 bits 16. (Overview.design, line 113).
- f0065 — PacketNumber has presence = required. (Overview.design, line 114).
- f0066 — PacketNumber has value-type = signed. (Overview.design, line 115).
- f0067 — PacketPayload has presence = required. (Overview.design, line 116).
- f0068 — PacketPayload has value-type = bytes. (Overview.design, line 117).
- f0069 — PersistentStore holds StoredData. (Overview.design, line 118).
- f0070 — Reader pursues Browse. (Overview.design, line 119).
- f0071 — Reader pursues Export. (Overview.design, line 120).
- f0072 — Remote consumes Api. (Overview.design, line 121).
- f0073 — Remote uses Bus as receiver of ChannelNotice in mode Online. (Overview.design, line 122).
- f0074 — Remote uses Bus as receiver of ChannelUpdate in mode Online. (Overview.design, line 123).
- f0075 — Remote uses Bus as sender of ChannelNotice in mode Offline. (Overview.design, line 124).
- f0076 — StoredData upholds DataRecord. (Overview.design, line 125).
- f0077 — UnusedShape has completeness = open. (Overview.design, line 126).
- f0078 — WireFamily from StoredData. (Overview.design, line 127).
- f0079 — WireFamily upholds WireShape. (Overview.design, line 128).
- f0080 — WireShape defines PacketBody. (Overview.design, line 129).

- f0081 — WireShape defines WireVariant. (Overview.design, line 130).
- f0082 — WireShape has completeness = closed. (Overview.design, line 131).
- f0083 — WireShape has-field PacketFlag. (Overview.design, line 132).
- f0084 — WireValue has presence = optional. (Overview.design, line 133).
- f0085 — WireValue has value-type = bytes. (Overview.design, line 134).
- f0086 — WireVariant has-field WireValue. (Overview.design, line 135).